MECHANICAL ENGINEER · ANKARA

# Mustafa Furkan Özel

Furkan is a mechanical engineer based in Ankara. He designs steel support structures for solar power plants and industrial buildings, works out their wind, snow and seismic loads, and turns them into fabrication-ready drawings. He is a Ph.D. student at Gazi University.

hi@furkanis.me ORCID Google Scholar

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## PROFILE

He focuses on handing the repetitive part of structural work to software. Instead of building SAP2000 models by hand, he writes Python scripts that generate them. His M.Sc. thesis applied the same idea to a buckling analysis framework that sweeps more than 20 variables across eight materials, published as a chapter with CRC Press.

Before that he led R&D on solar mounting systems under wind and snow loads at Fixus, ran FEA and CFD on TÜBİTAK-funded clean-energy projects at Theta, and advised on machining and cutting tools at Akifsan. His Ph.D. at Gazi University, advised by Prof. Olcay Ersel Canyurt, focuses on computational solid mechanics and design optimization.

## EXPERIENCE

3 yrs 10 mo in total

Jun 2026 — 6 mo

### Mechanical Engineer CURRENT

YBTB Mühendislik · Ankara

- Prepare steel structure designs, structural calculation reports and structural analyses for industrial and energy structures.
- Design steel substructures for ground-mount and rooftop solar projects in AutoCAD, SAP2000 and Fusion 360.
- Compile technical and commercial proposals for engineering packages.

Feb 2024 — Sep 2025
1 yr 8 mo

### Sales Engineer (Technical Consultant)

Akifsan Makina · Ankara

- Advised customers on machining processes, cutting tools and industrial equipment selection.
- Turned manufacturing constraints into technical specifications and wrote product documentation.

May 2023 — Feb 2024
10 mo

### R&D and Project Coordinator

Fixus Solar Mounting Systems · Ankara

- Led R&D on the structural optimization of solar mounting systems under wind and snow loads.
- Developed design alternatives that improved structural integrity and cut fabrication cost, with calculations reported to code.

Jan 2022 — Nov 2022
11 mo

### Design and Analysis Engineer

Theta Clean Energy Technologies · Ankara

- Ran CFD and structural FEA in ANSYS Fluent and ANSYS Mechanical on TÜBİTAK-funded clean-energy projects.
- Carried analysis models through to 3D CAD designs and physical prototypes.

## PUBLICATIONS

In press

**Parametric shear buckling analysis and multi-objective optimization of stiffened multi-cell box-beams**

Özel, M. F., & Girard, H. *Politeknik Dergisi / Journal of Polytechnic* (Gazi University; ESCI, TR Dizin). e-ISSN 2147-9429.

2025

**Shear buckling in thin-walled structures: Insights from Python for structural analysis**

Özel, M. F., & Girard, H. In H. Girard & H. Utku (Eds.), *Advances in Aerospace Engineering: Artificial Intelligence, Structures, Materials, and Optimization*, Ch. 3, pp. 35–46. CRC Press / Taylor & Francis. ISBN 978-1-003-37065-9.

## EDUCATION

Sep 2026 –

**Ph.D., Mechanical Engineering** ONGOING

Gazi University · Ankara

Advisor: Prof. Olcay Ersel Canyurt. Computational solid mechanics and design optimization.

Sep 2022 – Sep 2025

**M.Sc., Mechanical Engineering**

OSTİM Technical University · GPA 3.07/4.00

Thesis: Prevention of Shear Buckling in Thin-Walled Closed-Section Beams: Insights from Python-Based Structural Analysis. Advisor: Assoc. Prof. Hande Girard.

Aug 2016 – Jul 2021

**B.Sc., Mechanical Engineering**

Selçuk University · Konya · GPA 3.11/4.00

Graduated with honors.

Oct 2017 – Jun 2021

**B.A., Business Administration**

Anadolu University · Eskişehir · distance education

## SKILLS

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|---|---|
| SOFTWARE | SAP2000, ANSYS Mechanical, ANSYS Fluent, AutoCAD, Fusion 360, SolidWorks, MATLAB |
| PROGRAMMING | Python: parametric analysis, SAP2000 COM/OAPI automation, design optimization |
| LANGUAGES | Turkish (native) · English (CEFR B1; used for thesis and publications) |
| MEMBERSHIP | TMMOB Chamber of Mechanical Engineers (MMO) |

References available on request.